

```

from tensorflow.keras.datasets import fashion_mnist
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense
from tensorflow.keras.models import Sequential
import numpy as np
import matplotlib.pyplot as plt

(x_train, y_train), (x_test, y_test) = fashion_mnist.load_data()

print(x_train.shape, y_train.shape)

```

```
(60000, 28, 28) (60000,)
```

normalize

```

x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0

```

```

model = Sequential([
    Conv2D(
        filters=6,
        kernel_size=(2, 2),
        activation="relu",
        input_shape=(28,28, 1)
    ),
    MaxPooling2D(
        pool_size=(1,1)
    ),
    Flatten(),
    Dense(
        units=10,
        activation='softmax'
    )
])

```

```

model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
model.summary()

```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_2 (Conv2D)	(None, 27, 27, 6)	30
max_pooling2d_2 (MaxPooling2D)	(None, 27, 27, 6)	0
flatten_2 (Flatten)	(None, 4374)	0
dense_1 (Dense)	(None, 10)	43,750

Total params: 43,780 (171.02 KB)
Trainable params: 43,780 (171.02 KB)

```
histroy = model.fit(x_train, y_train, epochs=10, validation_data=(x_test, y_test), batch_size=32)
```

```

Epoch 1/10
1875/1875 ————— 26s 14ms/step - accuracy: 0.9199 - loss: 0.2170 - val_accuracy: 0.8576 - val_loss: 0.5007
Epoch 2/10
1875/1875 ————— 27s 14ms/step - accuracy: 0.9205 - loss: 0.2151 - val_accuracy: 0.8543 - val_loss: 0.5294
Epoch 3/10
1875/1875 ————— 27s 14ms/step - accuracy: 0.9199 - loss: 0.2137 - val_accuracy: 0.8629 - val_loss: 0.5442
Epoch 4/10
1875/1875 ————— 26s 14ms/step - accuracy: 0.9211 - loss: 0.2119 - val_accuracy: 0.8578 - val_loss: 0.5408
Epoch 5/10
1875/1875 ————— 25s 13ms/step - accuracy: 0.9256 - loss: 0.2022 - val_accuracy: 0.8592 - val_loss: 0.5371
Epoch 6/10
1875/1875 ————— 22s 12ms/step - accuracy: 0.9251 - loss: 0.2013 - val_accuracy: 0.8608 - val_loss: 0.5499
Epoch 7/10
1875/1875 ————— 23s 12ms/step - accuracy: 0.9249 - loss: 0.2015 - val_accuracy: 0.8399 - val_loss: 0.5867
Epoch 8/10
1875/1875 ————— 25s 13ms/step - accuracy: 0.9261 - loss: 0.1978 - val_accuracy: 0.8454 - val_loss: 0.6005
Epoch 9/10
1875/1875 ————— 25s 13ms/step - accuracy: 0.9270 - loss: 0.1930 - val_accuracy: 0.8496 - val_loss: 0.5859

```

Epoch 10/10

1875/1875 — 25s 14ms/step - accuracy: 0.9286 - loss: 0.1931 - val_accuracy: 0.8581 - val_loss: 0.5910

```
def plot_accuracy(history):  
    plt.figure(figsize=(10, 6))  
  
    plt.plot(history.history['accuracy'], label='Training Accuracy')  
    plt.plot(history.history['val_accuracy'], label='Validation Accuracy')  
  
    plt.xlabel('Epoch')  
    plt.ylabel('Accuracy')  
    plt.grid(True)  
    plt.title('Training and Validation Accuracy')  
    plt.legend(loc="upper right")  
    plt.show()  
  
plot_accuracy(histroy)
```

